CRYPTOGYAPHY AND NETWORK SECURITY ALA:3

Social Engineering Attacks and Major Security Breaches in India

A simple-English case-study report

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Academic note This report separates confirmed facts from allegations and does not treat an incident's attack method as social engineering unless the evidence supports that label.

Contents and reading guide

- Introduction and key attack techniques
- How a cyberattack commonly unfolds
- Five India-linked case studies
- Comparison table
- Prevention strategies
- Incident-response flowchart
- Conclusion and references

How to read the evidence

The five cases were selected because they are widely reported and useful for comparing different kinds of cyber incidents. A malware-based banking attack, a ransomware incident, a third-party data breach, an alleged

e-commerce exposure, and a disputed Aadhaar access story are not the same event. The report therefore uses careful labels such as confirmed, reported, potential, or allegation.

> **Source practice** Amounts, dates, affected systems and recovery statements are included only where a cited source reports them. “Not publicly confirmed” means the public record reviewed here did not establish a reliable figure.

1. Introduction

Cybersecurity is the practice of protecting computers, networks, applications and data from misuse, damage or unauthorized access. Social engineering is different from a purely technical attack: it targets people and their decisions. An attacker may pretend to be a bank employee, create urgency, offer a fake reward or ask someone to click a harmful link.

Many real incidents combine human and technical weaknesses. For example, a stolen password may give access to a system, and malware may then move through that system. The correct classification matters because prevention must address both technology and behaviour.

2. Common social-engineering and cyberattack techniques

Technique	Simple meaning and example
Phishing	A fake email, text or website tries to steal a password or payment detail. Example: “Your account will close today—sign in here.”
Spear phishing	A targeted message uses personal or workplace information to appear believable.
Pretexting	The attacker invents a story or identity, such as a fake auditor asking for an employee code.
Baiting	A tempting offer or free item encourages unsafe action, such as plugging in an unknown USB drive.

Quid pro quo	The attacker promises a benefit, service or refund in exchange for information.
Tailgating	A person follows an authorized employee into a restricted area.

Credential theft	Passwords, session tokens or one-time codes are captured or reused.
Malware	Malicious software changes, spies on or damages a system. It may arrive through a link, attachment or vulnerable service.
Ransomware	Malware locks or encrypts data and demands payment. It is a technical attack; social engineering may be one way it enters.
Software vulnerability	A flaw in an application or system is exploited. This is a technical weakness, not automatically a social-engineering attack.

3. How an attack commonly unfolds

Message or access point	Trust or mistake	Credentials / malware	Unauthorized access	Data or money stolen

This flow is a teaching model, not a claim that every incident followed the same sequence. In a technical breach, the first step may be an exposed service or vulnerable software rather than a message to a person.

Case Study 1: Cosmos Bank cyberattack (2018)

Evidence status Confirmed cyber-fraud reporting; exact full recovery amount and complete recovery duration were not publicly confirmed.

A. Case name and year

Cosmos Bank cyberattack — 2018.

B. History/background

Cosmos Bank, a Pune-based cooperative bank, reported a two-part cyber fraud in August 2018. Card data was used for coordinated ATM withdrawals in multiple countries, and a separate SWIFT-related transfer was sent to a Hong Kong-based account. The public reporting describes the event as a malware and payment-system compromise rather than a proven phishing attack.

C. Attack type and technique

ATM-switch malware, cloned-card transactions and fraudulent SWIFT transfer. This is primarily a technical and payment-fraud incident, not a confirmed social-engineering case.

D. How the attack happened

- Attackers compromised information connected to the bank's card-processing environment.
- They used stolen card data to create or use cloned cards.
- Thousands of cash withdrawals were made across many ATMs in a short period.

- A separate transfer was initiated through the bank's SWIFT environment to Hong Kong.
- Investigators traced part of the transferred money to Macau; later reporting described arrests and convictions



E. Original article photograph or official-report visual

[ORIGINAL ARTICLE VISUAL / SOURCE PLACEHOLDER]

A legally retrievable original news photograph was not embedded. The linked Reuters report includes a clearly labelled illustrative file image; the court-report link is provided for verification.

Reuters / The Hindu. "Cosmos Bank loses $13.5 million in cyber attack; Pune court convicts 11 accused." 2018-2023. Open source article/report

F. Impact

Reported losses were about ₹94.42 crore, including ₹80.50 crore through 14,849 card transactions and about ₹13.92 crore through the transfer route. The bank stated that customer accounts themselves were safe, but the bank and payment systems were affected.

G. Financial loss/cost

The reported principal loss was approximately ₹94.42 crore. Recovery of the full amount was not publicly confirmed in the sources reviewed; The Hindu reported that ₹5.72 crore of the Hong Kong transfer was recovered.

H. Recovery time

Cash withdrawals and transfer activity occurred over two days in August 2018. A full restoration duration for all systems was not publicly confirmed.

I. How the incident was discovered

The bank detected the unusual transactions and filed a police complaint; the case then became the subject of police investigation, RBI examination and court proceedings.

J. Response and solution

- The bank reassured customers and investigated the attack.
- Police investigated the ATM and SWIFT paths and arrested suspects.
- RBI officials examined the bank's IT robustness and the damage.
- A court later convicted 11 accused in the fraud case.

K. Prevention

- Segment ATM, card-switch and SWIFT environments.
- Use transaction velocity checks and geographic anomaly detection.
- Require independent approval for high-risk international transfers.
- Use hardware-backed keys, privileged-access monitoring and tested response playbooks.

L. Lessons learned

- Payment systems need controls that can stop abnormal speed and geography.
- A secure customer account does not mean the bank's processing layer is safe.
- International transfers need independent verification.
- Fast containment limits the value of a coordinated cash-out.

M. Sources

- Reuters — Cosmos Bank loses $13.5 million in cyber attack — https://www.reuters.com/article/world/cosmos-bank-loses-135-million-in-cyber-attack-idUSKBN1KZ1J8/
- The Hindu — Pune court convicts 11 accused in Cosmos Bank cyber fraud case — https://www.thehindu.com/news/national/other-states/pune-court-convicts-11-accused-in-cosmos-bank-cyber-fraud-case/a rticle66769928.ece
- Times of India — Cosmos Bank hit by cyber hack, loses Rs 94 crore in 2 days — https://timesofindia.indiatimes.com/city/pune/cosmos-bank-hit-by-cyber-hack-loses-rs-94-crore-in-2-days/articleshow/6540 9441.cms

Case Study 2: AIIMS Delhi ransomware incident (2022)

Evidence status Confirmed ransomware and service-disruption reporting; initial access and complete restoration date were not fully established in the cited sources.

A. Case name and year

AIIMS Delhi ransomware incident — 2022.

B. History/background

On 23 November 2022, the All India Institute of Medical Sciences (AIIMS), New Delhi experienced a ransomware attack that disrupted digital hospital operations. Government reporting said five servers were affected and about

1.3 terabytes of data were encrypted. Public sources describe improper network segmentation as a contributing weakness; the attacker's initial entry method was not publicly established in the cited government answer.

C. Attack type and technique

Ransomware and network-compromise incident. It should not be labelled phishing or social engineering without evidence of a deceptive message or human manipulation.

D. How the attack happened

- Threat actors gained access to parts of the AIIMS IT network.
- Poor network segmentation allowed the compromise to affect critical applications.
- Files and data on affected servers were encrypted.
- Hospital services such as registration, billing and laboratory workflows moved to manual processes.
- CERT-In and other stakeholders assessed the incident and advised remedial measures.

E. Original article photograph or official-report visual

[ORIGINAL ARTICLE VISUAL / SOURCE PLACEHOLDER]

The Hindu article includes an AP photograph, but it was not embedded here because the licensing status does not clearly permit reuse in a student assignment. The linked government PDF is the authoritative companion source.

The Hindu / Government of India. "1.3 TB data encrypted and five servers affected in AIIMS ransomware attack." 16-17 December 2022. Open source article/report

F. Impact

Five servers were affected and approximately 1.3 TB of data were encrypted, according to the Minister's written response reported by The Hindu. Patient-care operations continued, but many digital workflows were disrupted and manual queues increased.

G. Financial loss/cost

No reliable public total cost or ransom payment was confirmed in the sources reviewed. Reports stated that the government did not pay the ransom.

H. Recovery time

AIIMS restored its e-data after about six days according to Times of India, while online services and network sanitisation took longer. The precise date when every service was fully restored was not publicly confirmed in the reviewed sources.

I. How the incident was discovered

AIIMS informed authorities; an FIR was registered, CERT-In carried out an evaluation, and the incident was investigated by multiple agencies.

J. Response and solution

- The hospital shifted essential work to manual processes.
- CERT-In and other stakeholders evaluated the incident and issued remedial advice.
- Systems were sanitised and data was restored from available sources.
- Staff were instructed to install updated antivirus protection and follow new procedures.

K. Prevention

- Use strong network segmentation between clinical, administrative and backup systems.
- Maintain offline and immutable backups and test restoration.
- Apply least privilege, MFA and endpoint monitoring.
- Practice downtime procedures so patient care can continue safely.

L. Lessons learned

- Resilience is part of patient safety.
- Backups are useful only when restoration has been tested.
- Segmentation limits blast radius.
- Manual fallback procedures should be rehearsed before a crisis.

M. Sources

- The Hindu — 1.3 TB data encrypted and five servers affected in AIIMS ransomware attack — https://www.thehindu.com/news/national/13-tb-data-encrypted-in-ransomware-attack-on-aiims-by-unknown-threat-actors-c entre/article66271226.ece
- Government of India — Lok Sabha answer on cyber-attack in AIIMS — https://sansad.in/getFile/loksabhaquestions/annex/1710/AU1837.pdf
- Times of India — AIIMS restores data, online services to take time to resume — https://timesofindia.indiatimes.com/india/aiims-restores-data-online-services-to-take-time-to-resume/articleshow/95865392.cms



Probing server attack, CERT-In finds holes in AIIMS cyber security

Four servers infected, no measures were implemented by computer faculty to secure the network, finds agency

Written by Mahender Singh Manral Follow
New Delhi | December 4, 2022 1:18:22 am

NewsGuard

ADVERTISEMENT

The ransomware attack took place on AIIMS's servers on November 23. File

Probing into the cyberattack on some servers at the All India Institute of Medical Sciences (AIIMS) in Delhi, the Indian Computer Emergency Response Team (CERT-In) has found that no measures were implemented by AIIMS computer faculty for securing the network; that no policies were defined on their firewall and in their network; and most of the switches were unmanaged.

LIVE BLOG

- France vs Poland, FIFA World Cup 2022 Live Updates: FRA cross swords with POL with QF spot at stake — 52 mins ago
- Delhi MCD Elections Live Updates: 50% voter turnout till 5.30 pm; as air quality turns 'severe', construction and demolition ban kicks in — 54 mins ago
- Gujarat Assembly Elections Live Updates: Voting for Phase 2 on December 5; BJP, Congress exude confidence in party victory — 5 hours ago
- FIFA World Cup 2022 Day 15 Live Updates: France face Poland, England go up against Senegal — 10 hours ago

Case Study 3: Air India passenger data breach (2021)

Evidence status Confirmed public disclosure of a third-party cyberattack affecting Air India passenger records; attacker identity and recovery cost were not confirmed.

A. Case name and year

Air India passenger data breach — 2021.

B. History/background

Air India disclosed in May 2021 that a cyberattack at SITA, its passenger service system provider, affected approximately 4.5 million passengers worldwide. The affected records covered passengers registered over a long period, including personal, passport, ticket and frequent-flyer information. Air India said CVV/CVC numbers were not held by the processor.

C. Attack type and technique

Third-party supplier compromise / data breach. The public reporting does not establish phishing, pretexting or another social-engineering technique as the cause.

D. How the attack happened

- SITA, a data processor used by airlines, suffered a cyberattack in February 2021.
- Air India was notified and later received more detail about the affected data.
- Passenger records from the relevant historical period were identified as exposed.
- Air India secured affected servers, investigated with external specialists and advised password changes.

E. Original article photograph or official-report visual

[ORIGINAL ARTICLE VISUAL / SOURCE PLACEHOLDER]

The linked reports include publisher-owned news images or file photographs, but they were not embedded because reuse rights were not clearly available. Source links and captions are retained.

Reuters / BBC News. "Air India says February's data breach affected 4.5 mln passengers." 21-22 May 2021. Open source article/report

F. Impact

Names, dates of birth, contact details, passport information, ticket information and frequent-flyer data were involved. Air India said no password data was affected and that CVV/CVC numbers were not stored by SITA.

G. Financial loss/cost

Not publicly confirmed. The public reports reviewed describe the breach and response but do not establish a verified Air India recovery-cost total.

H. Recovery time

The incident was first reported to Air India by SITA in February 2021; the full scope was communicated in stages. A complete technical recovery duration was not publicly confirmed.

I. How the incident was discovered

SITA notified Air India about the cyberattack. Air India investigated and issued a public notification in May 2021.

J. Response and solution

- Air India secured compromised servers and engaged external cybersecurity specialists.
- The airline notified affected passengers and encouraged password changes.
- The airline continued to investigate the third-party incident and data scope.

K. Prevention

- Perform supplier security assessments and require rapid incident notification.
- Minimise data stored by processors and separate sensitive fields.
- Use MFA, strong password policies and monitoring for customer accounts.
- Maintain a clear third-party breach communications plan.

L. Lessons learned

- A supplier can become the route to a major customer-data incident.

- Data minimisation reduces harm.

- Breach notification may occur in stages as investigations mature.
- Passengers need clear, practical protective advice, not only a technical explanation.

M. Sources

- Reuters — Air India says February's data breach affected 4.5 mln passengers — https://www.reuters.com/world/india/air-india-says-februarys-data-breach-affected-45-mln-passengers-2021-05-21/
- BBC News — Air India cyber-attack: Data of millions of customers compromised — https://www.bbc.com/news/world-asia-india-57210118
- TechCrunch — Air India passenger data breach reveals SITA hack — https://techcrunch.com/2021/05/23/air-india-passenger-data-breach-reveals-sita-hack-worse-than-first-thought/



Case Study 4: BigBasket potential data breach (2020)

Evidence status Potential or alleged exposure; company was evaluating authenticity and extent. It is not presented as a fully proven breach of all claimed records.

A. Case name and year

BigBasket potential data breach — 2020.

B. History/background

In November 2020, cybersecurity firm Cyble reported that a database allegedly belonging to BigBasket had been offered for sale on a cybercrime market. The company said it was evaluating the claim, checking authenticity and had filed a police complaint. This case is included as a documented potential exposure, not as an unqualified proven breach of every claimed record.

C. Attack type and technique

Potential database exposure / credential and personal-data risk. The public sources reviewed do not prove that phishing, pretexting or baiting caused the incident.

D. How the attack happened

- Cyble found a database offered in a cybercrime market and investigated the sample.
- The firm reported that the data appeared to include names, emails, phone numbers, addresses, dates of birth and password hashes.
- Cyble informed BigBasket and later publicised the finding.
- BigBasket filed a complaint and began evaluating the authenticity and extent of the claim.

E. Original article photograph or official-report visual

[ORIGINAL ARTICLE VISUAL / SOURCE PLACEHOLDER]

No original article photo was embedded because the reports use file or stock imagery and reuse rights were not clearly available. The source links document the claim and the company response.

Business Standard / The Indian Express. "BigBasket faces potential data breach; details of 20 mn users put on sale." 8-12 November 2020. Open source article/report

F. Impact

Cyble claimed that data for close to 20 million users was involved. The company said it did not store financial data such as credit-card numbers. Because the authenticity and full scope were under investigation, the report should be read as a potential exposure.

G. Financial loss/cost

The alleged data was advertised for about US$40,000 or approximately ₹30 lakh, but that is the seller's asking price, not BigBasket's loss. A verified company loss or recovery cost was not publicly confirmed.

H. Recovery time

Not publicly confirmed. The public sources describe investigation and containment efforts, not a verified restoration period.

I. How the incident was discovered

Cyble detected and validated the sample before notifying BigBasket; the company then filed a complaint and investigated the claim.

J. Response and solution

- BigBasket filed a cybercrime complaint.
- The company consulted cybersecurity experts and evaluated the authenticity and extent.
- The company stated that it did not store financial card data and sought immediate containment.

K. Prevention

- Protect password hashes with modern, slow password hashing and salt.
- Monitor for leaked credentials and force resets when exposure is credible.
- Limit personal data collection and retention.
- Use database access monitoring, segmentation and tested incident-response procedures.

L. Lessons learned

- A reported leak needs verification before its scale is treated as fact.
- Asking prices on criminal markets are not the victim's financial loss.
- Password hygiene still matters when passwords are hashed.
- Companies should communicate what is known, unknown and being investigated.

M. Sources

- Business Standard — BigBasket faces potential data breach — https://www.business-standard.com/article/companies/bigbasket-database-of-20-million-customers-hacked-12011080080_0_1.html
- The Indian Express — Explained: How big is the Bigbasket data breach? — https://indianexpress.com/article/explained/explained-how-big-is-the-bigbasket-data-breach-7026688/
- The Hindu — Bigbasket faces potential data breach — https://www.thehindu.com/business/Industry/bigbasket-faces-potential-data-breach-details-of-2-crore-users-put-on-sale-on-dark-web/article33056301.ece

Data breach at BigBasket hits 20 mn users' privacy

MONDAY, 9 NOVEMBER 2020
MUMBAI
mint

Leak contains names, email IDs, password hashes, pin, contact numbers, addresses

Sharan Poovanna
sharan.p@livemint.com
BENGALURU



Breach occurred on 14 Oct and a US-based cybersecurity firm detected irregularities a fortnight later. MINT

Online grocer Bigbasket may have suffered a massive data breach following which details of more than 20 million users may have been leaked on the dark web, said a US-based cybersecurity firm. Data worth $40,000 (₹30 lakh) was sold, the Atlanta-headquartered Cyble Inc's research team found during routine dark web monitoring, it said in a blogpost on Saturday.

"The leak contains a database portion with the table name 'member_member'. The size of the SQL file is ~15 GB, containing close to 20 million user data. More specifically, this includes full names, email IDs, password hashes (potentially hashed OTPs), pin, contact numbers (mobile plus phone), full addresses, date of birth, location, and IP addresses of login among many others," said Cyble.

The alleged breach occurred on 14 October and Cyble detected the irregularities a fortnight later, according to the blogspot.

It was validated and shared on 1 November with the management of the Bengaluru-headquartered BigBasket, which has filed a case with the cyber crime cell in the city.

Bigbasket.com, which is run by Innovative Retail Concepts Pvt Ltd, is one of India's largest online food and grocery store and is valued at $2 billion. The company is backed by Alibaba Group and Mirae Asset-Naver Asia Growth Fund, among others.

Bigbasket is evaluating the extent of the breach and authenticity of the claim with cybersecurity experts, besides finding ways to contain it, the online grocer said. "The privacy and confidentiality of our customers is our priority and we do not store any financial data, including credit card numbers, and are confident that this financial data is secure. The only customer data that we maintain are email ids, phone numbers, order details, and addresses so these are the details that could potentially have been accessed," the company said.

Bigbasket has a robust information security framework that employs best-in-class resources and technologies to manage information, it said. "We will continue to proactively engage with best-in-class information security experts to strengthen this further," it said.

The online retailer has more than 18,000 products and 1,000 brands in its catalogue and services customers in more than 20 cities across India.

The breach comes at a time when the lockdown to contain the spread of coronavirus encouraged more people to shop online for essentials such as grocery. However, online activity increased the threat of phishing attacks, though there has been no sharp spike in cyber crime activity, Bengaluru police said.

In June, delivery startup Dunzo had experienced a data breach in which the personal details of more than 300,000 accounts were leaked.

CYBERATTACK DETAILS

DATA worth ₹30 lakh was sold, Cyble Inc's research team found during routine dark web monitoring

BREACH details were disclosed to BigBasket on 1 Nov, which filed a case in Bengaluru cyber cell

BIGBASKET is now evaluating the extent of the breach and authenticity of the claim with experts

THE only customer data we maintain are email ids, numbers, order details, and addresses: BigBasket

Case Study 5: Aadhaar data-exposure allegations (2018)

Evidence status Disputed allegation. UIDAI denied a biometric database breach; The Tribune and other reporting described unauthorized access to demographic lookup functionality.

A. Case name and year

Aadhaar data-exposure allegations — 2018.

B. History/background

In January 2018, The Tribune reported that an intermediary allegedly sold access to a search facility that returned Aadhaar-related demographic information. UIDAI denied that the core Aadhaar biometric database had been breached, described the incident as misuse of a grievance-redressal facility and said it had logging and traceability. This is therefore an allegation and access-control controversy, not a confirmed biometric-database breach.

C. Attack type and technique

Alleged unauthorized use of a grievance-redressal access facility and credential sharing. It is not accurate to call the event a proven breach of the biometric database.

D. How the attack happened

- The Tribune said a reporter paid an intermediary and received login details for a search facility.
- The report said demographic details could be retrieved using an Aadhaar number.
- UIDAI said the facility was limited, logged and intended for authorized personnel.
- UIDAI denied a biometric database breach and said legal action was being pursued.

E. Original article photograph or official-report visual

[ORIGINAL ARTICLE VISUAL / SOURCE PLACEHOLDER]

The linked BBC article credits a Getty image and the PIB release is text-only; no image was embedded because a reusable original visual was not legally clear. The disagreement is documented explicitly.

PIB / BBC News. "UIDAI denies media report and says Aadhaar data was not breached." 4-5 January 2018. Open source article/report

F. Impact

The reported exposure concerned demographic information such as name and address; the sources disagree about the characterisation of the incident. UIDAI stated that biometric data remained safe and that the facility did not provide biometric access.

G. Financial loss/cost

Not publicly confirmed. The ₹500 payment described in reporting was an alleged access fee, not a verified loss suffered by UIDAI or citizens.

H. Recovery time

Not publicly confirmed. The sources reviewed discuss investigation and legal action, not a verified restoration period.

I. How the incident was discovered

The newspaper's investigation brought the story to light. UIDAI issued a public denial, said it maintained logs and traceability, and reported that an FIR/legal action was being pursued.

J. Response and solution

- UIDAI denied a core biometric database breach and explained the intended purpose and limits of the facility.
- UIDAI said logs could support traceability and legal action.
- The Tribune defended its reporting and responded to UIDAI queries.

K. Prevention

- Use strict role-based access and individual accounts; never share credentials.
- Log every query, alert on unusual volume and review access regularly.
- Separate demographic lookup from high-risk identity operations.
- Use MFA, approvals and rapid suspension for compromised accounts.

L. Lessons learned

- A public dispute is not proof of a full database breach.
- Limited interfaces can still create privacy risk if access is misused.
- Audit logs are valuable only when reviewed and acted on.
- Privacy claims need clear scope: demographic data and biometric data are not the same.

M. Sources

- PIB — UIDAI denies media report and says Aadhaar data was not breached — https://www.pib.gov.in/PressReleasePage.aspx?PRID=1515503
- BBC News — Aadhaar: Leak in world's biggest database worries Indians — https://www.bbc.co.uk/news/world-asia-india-42575443
- The Tribune — Tribune response to UIDAI email on Aadhaar expose — https://www.tribuneindia.com/news/archive/nation/tribune-response-to-uidai-email-on-aadhaar-expose-525479/

6. Comparison of the five cases

Case / year	Attack type	Main impact	Reported financial loss	Recovery time	Key lesson
Cosmos Bank / 2018	ATM-switch malware; cloned cards; SWIFT fraud	Bank payment systems and cash withdrawals	₹94.42 crore reported principal; full recovery not confirmed	Two-day attack; full restoration not confirmed	Monitor payment anomalies and segregate critical systems
AIIMS Delhi / 2022	Ransomware; network compromise	Five servers; ~1.3 TB encrypted; hospital services disrupted	Not publicly confirmed; government said ransom not paid	E-data restored after about six days; full service date not confirmed	Segment networks and test offline backups
Air India / 2021	Third-party supplier cyberattack	~4.5 million passenger records affected	Not publicly confirmed	Scope disclosed in stages; full technical recovery not confirmed	Manage supplier risk and minimise stored data

BigBasket / 2020	Potential database exposure	Claimed ~20 million records; authenticity under review	Seller asking price US$40,000 / ~₹30 lakh; not victim loss	Not publicly confirmed	Verify claims and protect password data

Aadhaar / 2018	Alleged misuse of grievance lookup access	Demographic-data access allegation; biometric breach denied	Not publicly confirmed	Not publicly confirmed	Separate disputed claims from confirmed facts

7. Prevention strategies

Control	Practical action
Multi-factor authentication	Use a second factor for email, banking, admin and remote access. Do not approve an unexpected login prompt.
Strong, unique passwords	Use a password manager and never reuse a password across services.
Software updates	Patch operating systems, applications, network devices and security tools quickly.
Access controls	Give people only the access they need. Use separate admin accounts and review permissions.
Employee awareness	Train staff to pause, verify unusual requests and report suspicious messages without fear.
Secure backups	Keep offline or immutable backups. Test restoration, not just backup creation.
Monitoring	Alert on unusual logins, locations, transaction speed, data downloads and privilege changes.
Incident response plan	Define who isolates systems, contacts authorities, communicates with customers and restores services.
Prompt reporting	Report suspicious messages, lost devices, account compromise and unusual transactions immediately.

8. Incident-response flowchart

Detect	Contain	Investigate	Remove threat	Restore systems	Monitor and improve

The goal is to stop further damage while preserving evidence, then restore trusted systems and improve controls. Organisations should document decisions and communicate clearly during each stage.

9. Conclusion

The five cases show that “cyberattack” is a broad term. Cosmos Bank involved coordinated payment fraud, AIIMS involved ransomware and operational disruption, Air India’s incident began at a service provider, BigBasket was reported as a potential exposure, and the Aadhaar story remained disputed between investigative reporting and UIDAI’s denial of a biometric breach. A good cybersecurity programme therefore combines technical controls, human awareness, supplier governance, privacy protection and a practiced response plan.

The most practical lesson for an IT student is to avoid one-size-fits-all labels. First identify what was actually reported, then ask which weakness allowed it: a person, a password, a supplier, an exposed service, weak segmentation or poor monitoring. Prevention becomes more effective when the control matches the weakness.

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   https://www.reuters.com/article/world/cosmos-bank-loses-135-million-in-cyber-attack-idUSKBN1KZ1J8/

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Appendix: evidence labels used in this assignment

- Confirmed reporting: the cited source directly reports the event or an official statement.
- Reported amount: a source gives the figure, but it may not equal the victim's final loss or recovery cost.
- Potential / alleged: a credible report describes a claim that the organisation or authorities were still verifying.
- Disputed: different parties publicly disagreed about what happened or how it should be described.
- Not publicly confirmed: no reliable figure or duration was established in the reviewed sources.

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CONSTITUTION BENCH UPHOLDS AADHAAR LAW 4-1, WANTS STRONG FIREWALLS

Supreme Court gives Aadhaar some privacy

Fair balance between privacy and right to life: CJI bench

ANANTHAKRISHNAN G
NEW DELHI, SEPTEMBER 26

...AI and marks majority ruling on the world's largest biometric identity type ject, the Supreme Court Wednesday upheld the Constitutional validity of the ...dhaar Act on the ground that ... "struck a fair balance between ...ht right of privacy of the ...dividual with right to life of the ...ne individual as a beneficiary" ...ot ring-fenced the law from ...sable misuse by the State and ...rate entities.

What it upheld the linking of ...e permanent account number (PAN) to the unique Aadhaar ...mber and the notification ...lowing people to avail ...subsidies and benefits provided ...y the government, a five-judge Constitution Bench, in a 4-1 ...rdict on several ...petitions, ...truck down the requirement of ...nking Aadhaar to bank ac...ounts and mobile phone num...ers, and on its insistence for ...ension and school admissions.

It expressed the opinion that ...he government "shall not unduly expand the scope of 'subsidies, services and benefits' ...hereby widening the net of ...adhaar, where it is not permitted otherwise".

CONTINUED ON PAGE 2



Ranjana Sonawane, the first Aadhaar beneficiary, at her home in Tembhli village, Nandurbar, Maharashtra, Wednesday. She got her card in 2010 and, apart from monthly rations, used it to get a subsidised gas cylinder, scholarship for her son. *Prabhat Nafde* REPORT, PAGE 9

BUSINESS AS USUAL
BY UNNY



14 THE EDITORIAL PAGE

Course correction
BY ARVIND PADTAR

STRIKES DOWN PART OF SECTION 57

Private firms (banks and phones) can't ask for Aadhaar linking

KRISHN KAUSHIK
NEW DELHI, SEPTEMBER 26

IN UPHOLDING the Constitutional validity of the Aadhaar programme in a 4-1 ruling, a Constitution Bench of the Supreme Court also limited the scope of the biometric project, particularly on sharing data with private entities and ruled that Aadhaar was no longer mandatory for bank accounts, mobile connections, pension or school admissions.

Shaping digital destiny

Aadhaar verdict will cleanse system of fakes and duplicates, ensure services reach intended beneficiaries.
THE EDITORIAL

The majority verdict by Chief Justice Dipak Misra and Justices A K Sikri, who wrote the majority judgment, and A M Khanwilkar broadly chastising the Centre for trying to push Aadhaar on children had a range of suggestions to strengthen the Aadhaar (Targeted Delivery of Financial and other Subsidies, Benefits and Services) Act, 2016. It also barred corporate bodies from using Aadhaar for authentication unless backed by a law and ruled that the UIDAI cannot store any metadata related to Aadhaar-based authentications or authentication history for

CONTINUED ON PAGE 2

'DID NOT QUALIFY AS MONEY BILL'

ECOSYSTEM HIT

Govt thumbs up but fingers crossed in tech space

KARISHMA MEHROTRA & RAVISH TIWARI
NEW DELHI, SEPTEMBER 26

WHILE THE government has reason to cheer the Supreme Court verdict, there is uncertainty and anxiety in the nascent digital ecosystem building around the Aadhaar digital infrastructure.

Beyond the declared purpose for targeted welfare/subsidy delivery, Aadhaar has grown as a digital infrastructure created by

MoD official put on record objections to 36-Rafale deal

Joint Secy (Air) objected to benchmark price, was overruled; Cabinet approved in September 2016; officer went on leave



SUSHANT SINGH
NEW DELHI, SEPTEMBER 26

NEARLY A month before the deal for 36 Rafale aircraft was signed between then Defence Minister Manohar Parrikar and his French counterpart in New Delhi in September 2016, a senior officer of the Ministry of Defence, who had been part of the Contract Negotiations Committee (CNC), raised questions about the deal's benchmark price and put his objections on record, *The Indian Express* has learnt.

This officer was then Joint Secretary & Acquisition Manager (Air) in the MoD and the one empowered to initiate the note for the Cabinet's approval.

Sources have confirmed to *The Indian Express* that his objections delayed the Cabinet note approving the deal and its signing which only happened after his objections were "overruled" by another senior MoD official, Director General (Acquisition).

The note by the JS&AM (Air), as part of the Rafale deal file, is currently with the office of the Comptroller and Auditor General (CAG) who is studying the deal which has set off a political controversy.

Sources said the CAG is likely

CONTINUED ON PAGE 2

It was a govt-to-govt discussion... I was not in charge at that time... We have very clear rules... This is very important to me because this is a strategic coalition

EMMANUEL MACRON, FRENCH PRESIDENT

RELATED REPORT PAGE 10

Quota in promotions OK but keep out creamy layer among SCs, STs: Bench

ANANT HAKRISHNAN G & SEEMA CHISHTI
NEW DELHI, SEPTEMBER 26

APPLYING THE creamy layer principle to promotions for Scheduled Castes and Scheduled Tribes in government jobs, the Supreme Court Wednesday declined to refer to a larger bench its 2006 verdict on the matter. It ruled that states no longer need to collect quantifiable data on the backwardness of SCs and STs in granting reservation in promotions, but will have to back it with data to show that inadequate representation in the order.

Ruling unanimously on a batch of petitions to revisit the *M Nagaraj & Others vs Union of India* verdict, a five-judge Constitution Bench, comprising Chief Justice of India Dipak Misra, Justices Rohinton Fali Nariman (he wrote the order), S K Kaul and Indu Malhotra, said: "The whole object of reservation is to see that backward classes of citizens move forward so that they may march hand in hand with other citizens of India on an equal basis. This will not be possible if only the creamy layer

CONTINUED ON PAGE 2

On EC table: Model code to kick in as soon as House dissolved

P V AJAYA NATHAN IYER & RITIK KAPOOR
NEW DELHI, SEPTEMBER 26

THE ELECTION Commission is considering passing an order shortly to ensure that the Model Code of Conduct kicks in immediately after a state dissolves its Assembly and seeks early elections. This will mean that states may be barred from taking any policy decision which may have an influencing the electorate. The kicking in of the Model Code of Conduct may not

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